Aniket Khan

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Education

Program	Institution	%/CGPA	Completion
B.Tech - Mechanical Engineering	Indian Institute of Technology, Madras	8.93/10.0	2025
Minor - **AI & Machine Learning**	Indian Institute of Technology, Madras	9.5/10.0	2025
Class XII, CBSE	D.A.V. Public School, Kolkata	97.0%	2021
Class X, CBSE	D.A.V. Public School, Kolkata	96.2%	2019

Scholastic Achievements

- Awarded the prestigious **National Talent Search Examination (NTSE)** scholarship (2019)
- Appeared in **Regional Mathematics Olympiad (RMO)** from West Bengal (2019)
- Appeared in the interview round of **Kishore Vaigyanik Protsahan Yojana (KVPY)** in SA stream (2020)

Skills and Online Courses

- **AI/ML:** Deep Learning (PyTorch, scikit-learn, PyTorch Geometric), Reinforcement Learning (Gym/Gymnasium, Stable-Baselines3), Transformers & NLP (Word2Vec, BERT, SBERT), Quantum Machine Learning (Qiskit)
- **Programming:** Python, C/C++, git, MATLAB
- **Robotics/Simulation Tools:** ROS, Gazebo, RViz, Arduino, Simulink, Ansys Fluent (scripting)
- **Other Tools:** AutoCAD, Fusion 360, Onshape, Tinkercad
- **Competitions:** TechSoc IoT (Runner-up) — designed smart embedded systems integrating ESP32, sensors, and APIs (Bluetooth door lock, adaptive fan control)

Research and Professional Experience

Computational Neuroscience Lab (Research Intern) — May 2024 – July 2025

(Department of Biotechnology, IIT Madras)

RL for Nutrition-Aware Financial Decision Making — July 2024 – Oct 2024

- Trained a reinforcement learning agent to optimize food choices by balancing **cost vs. long-term nutritional value**
- Designed reward shaping and environment logic to **encourage sustainable decision-making** over time

Multi-Agent RL: Brain–Gut Coordination — Nov 2024 – July 2025

- Built a **multi-agent RL (MARL)** setup involving discrete and continuous agents simulating human physiology
- Brain agent (discrete) decides to forage or eat; Gut agent (continuous **DDPG**) allocates nutrients post-consumption
- Modeled **inter-organ interaction** through shared rewards and realistic physiological energy dynamics

Graph-based Classification of Schizophrenic EEG Data — May 2024 – July 2024

- Automated **data preprocessing pipeline** for EEG-3 signals — filtering, correlation, PSD, and batching of inputs
- Built a **Graph Convolutional Neural Network (GCNN)** for classifying schizophrenic vs. healthy individuals (N=63)
- Applied **round-robin cross-validation** to capture inter-region neural correlations in brain cluster networks

Bharat Petroleum Corporation Ltd. — June 2025 – Present

(Assistant Executive – Inspection, Engineering & Advisory Services at BPCL, a Fortune 500 public sector oil & gas company)

- Responsible for equipment reliability, statutory compliance, and corrosion monitoring at Mumbai Refinery
- Completed **2 months of field training** across BPCL Business Units in operations, maintenance, and safety

Thryv Mobility (Intern) — June 2023 – Feb 2024

(Startup working on intelligent mobility solutions)

Development of a Self-Balancing Smart Wheelchair

- Implemented **Proximal Policy Optimization (PPO)** on Gazebo-ROS simulation for **yaw stability** on uneven terrains
- Experimented with **traditional + learning-based controllers** to stabilize inverted pendulum-style wheelchair model
- Fabricated a **physical prototype** with modular weights, custom-built controller mountings, and sensor housings

BioNex Lab (Undergraduate Research Project) — Jan 2024 – June 2024

(Guide: Dr. Manish Anand, Department of Engineering Design, IIT Madras)

- Built an **image recognition-based wheelchair detector** using **YOLOv8** on real-time video feed
- Annotated and preprocessed a custom training dataset with 500+ images from open-source sources
- Used 3 class **confusion matrix** for evaluation; achieved **position estimates within ±2cm** uncertainty

Heat Transfer & Thermal Power Lab (Undergraduate Research Project) July 2024 – Nov-2024

(Guide: Dr. Chakravarthy Balaji, Department of Mechanical Engineering, IIT Madras)

- Developed and validated a **numerical model for nucleate boiling** of dielectric fluid (Novec 649) in Ansys Fluent
- Implemented the **k-Ishii** model for **Bubble Departure Diameter**, achieving highest accuracy against expt data
- Optimized mesh quality, performed **grid independence study**, and explored UDF-based boiling models

Select Course Projects

Reinforcement Learning Algorithms Jan 2024 – May 2024

- Implemented **Q-Learning & SARSA**, & studied their performance on a **custom-built Gridworld environment**
- Developed **Dueling Deep Q Network (DDQN)** and **Monte Carlo REINFORCE** for **Acrobot and CartPole**
- Built **SMDP and Intra-option RL** on a **Taxi environment** using custom options and analyzed agent behavior

Quantum Error Mitigation using ML Jan 2025 – May 2025

- Implemented **ML-based quantum error mitigation** (GCN, GAT, KAN) and benchmarked against **ZNE techniques**
- Encoded noisy quantum circuit outputs into graph, frequency, and CNN formats for ML training
- Trained and tested models using both **quantum simulators** and **IBM Q hardware** data
- Achieved lower **L2 error and MAE**; ML models showed **faster runtime** than traditional ZNE

Modular NLP-based Information Retrieval System Jan 2025 – May 2025

- Developed a modular IR system combining **TF-IDF**, **LSA**, **autoencoders**, and **S-BERT** for semantic retrieval
- Integrated **query expansion**, **word sense disambiguation** (Lesk variants), and **spell correction** for preprocessing
- Evaluated on Cranfield dataset using **Precision**, **Recall**, **F1**, **MAP**, and **nDCG**; **S-BERT** outperformed all baselines

Parallel 2D Discrete Fourier Transform (DFT) Jan 2024 – May 2024

- Implemented **2D DFT and inverse DFT from scratch** in C/C++ using **OpenMP** and **MPI**
- Evaluated performance of the parallel scheme by computing **scalability, efficiency, and speedup** of the code
- Achieved **over 80% efficiency** with near-linear speedup and a **1:1 correspondence** between threads and speedup

Relevant Courses

- Reinforcement Learning
- Foundations of Machine Learning
- Machine Learning Techniques
- Natural Language Processing
- Quantum Computing and Machine Learning
- Deep Learning (NPTEL, Govt. of India)
- Parallel Scientific Computing
- Linear Algebra for Engineers
- Statistical Design and Analysis of Experiments
- Modern Control Theory
- Design and Optimization of Energy Systems
- Product Reliability
- Responsible and Safe AI(ongoing, NPTEL)
- Introduction to German

Co-Curricular and Extra-Curricular Activities

Schroeter and Dean's Trophy 2023 *(Inter-Hostel Sports Event)*

- Secured **Gold Medal** while representing Cauvery Hostel in a round-robin tournament of **Water Polo**
- Secured double victory by winning **2 Gold medals** in the swimming events: **200m Medley Relay** and **400m Relay**
- Contributed to a remarkable team achievement by winning the **silver medal** in the **table tennis** tournament
- Finished Dean's Trophy **triathlon**(500m swimming, 7.8km cycling, and 3.3km running) in the **top 20**

Others

- Successfully finished training in swimming by **National Sports Organisation(NSO)** in 2022
- Nominated as a **potential member** of the prestigious Madras Sharks aquatics team for **inter-IIT sports meet**
- Secured the **runner-up** position at the **table tennis** tournament organized by the MEA during sports weekend 2023